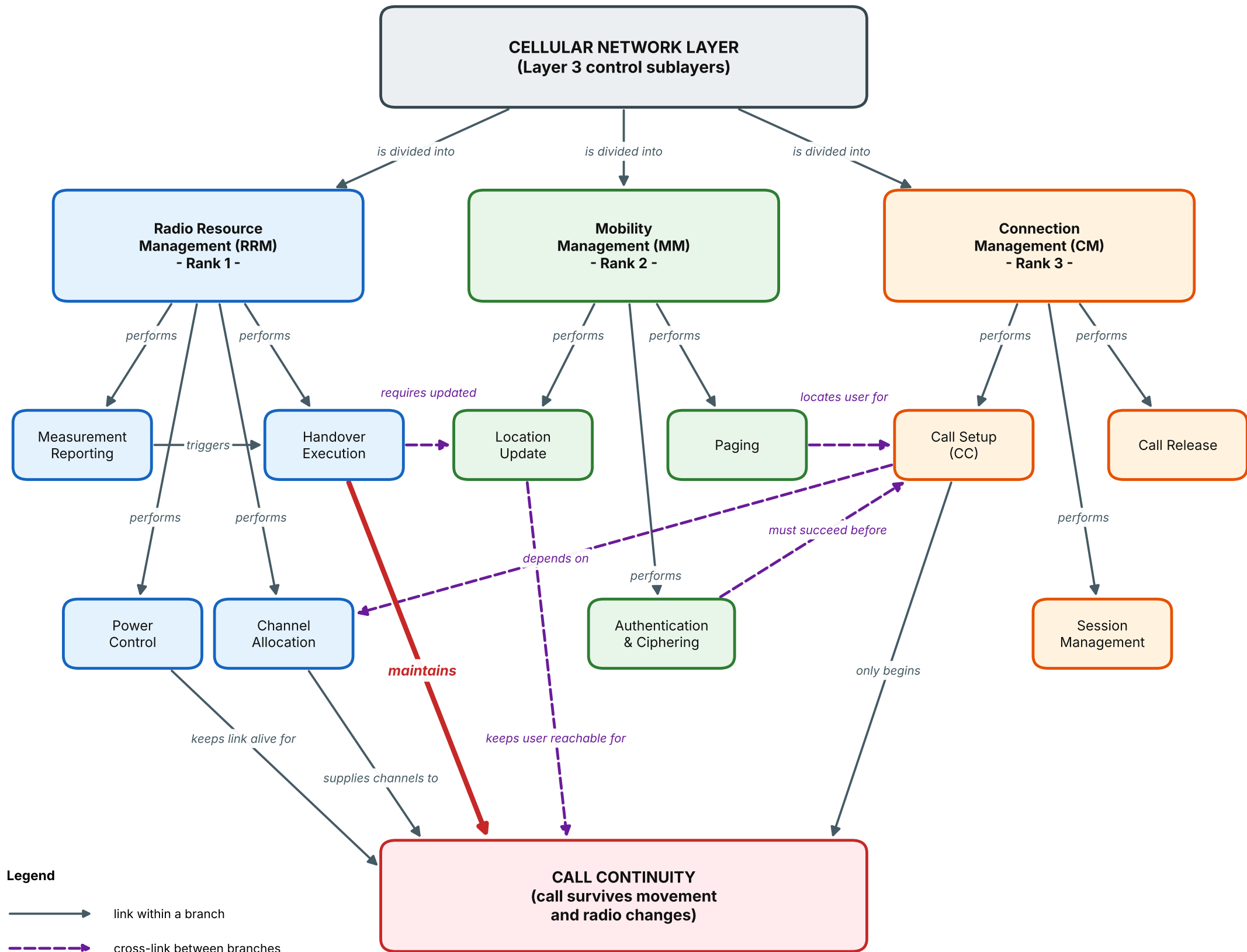


Concept Map: Evaluating the Sublayers of the Cellular Network Layer

Focus question: Which sublayer has the largest effect on call continuity?



Evaluation and Justification

Ranking: RRM (1st) > MM (2nd) > CM (3rd). RRM ranks first because its procedures work while a call is in progress: measurement reporting triggers handover execution, while power control and channel allocation keep the radio link usable. If they fail, the call drops immediately. MM ranks second: location update and paging keep the network aware of where the user is, and authentication admits the user, but these mostly act before the call or support a handover. CM ranks third because call setup and release only mark the start and end of a call and do nothing while it runs. I chose "triggers", "maintains" and "must succeed before" because they show cause, effect and order, not just association. "Depends on" and "requires updated" show that the sublayers rely on each other, so the cross-links reveal cooperation. Only RRM's handover has a direct "maintains" link to call continuity. Conclusion: Radio Resource Management has the largest effect on call continuity, Mobility Management has a supporting effect, and Connection Management has the smallest.

Real-World Example: Handover in GSM and 4G (LTE)

In GSM, the three Layer-3 sublayers are exactly RR, MM and CM (3GPP TS 44.018 / 24.008). During a call, the phone measures its serving and neighbouring cells and reports them to the network on the SACCH roughly every 480 ms. The base station controller compares the reports with its thresholds, decides on a handover and sends a HANDOVER COMMAND, so the call moves to a new cell with only a very short break. In 4G LTE the same RRM role is played by measurement reports (for example event A3, where a neighbour cell becomes better than the serving cell) followed by an X2 or S1 handover. The call survives only because of this RRM procedure; location update, authentication and call setup were already finished before the handover began, which supports the ranking above.